Governance of Explainable AI in Internal Financial Control Systems: A Proposed Framework Integrating XAI Techniques with the COSO Model to Enhance Institutional Sustainability

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Abstract

Machine-learning models are increasingly embedded in internal financial control, yet their opacity conflicts with the transparency and accountability that control frameworks require. This study proposes IFC-XAI, a governance framework that links explainable artificial intelligence (XAI) outputs to the five components of the COSO Internal Control–Integrated Framework (2013), and examines its feasibility empirically. Following a Design Science Research approach, four classifiers (logistic regression, decision tree, random forest and gradient boosting) were trained on the public German Credit dataset (n = 1000; 20 raw and 61 encoded features). On a stratified hold-out test set (n = 300), logistic regression achieved the highest area under the ROC curve (AUC = 0.801), followed by random forest (0.797); under repeated stratified cross-validation (5 folds × 3 repeats) the two models were statistically indistinguishable (paired t-test, p = 0.058). SHAP attributions for the logistic regression model were aggregated by COSO component: Risk Assessment accounted for 55.9% of the explanatory weight of the top 15 features, Control Activities for 28.2% and Control Environment for 15.9%. Lowering the decision threshold from 0.50 to 0.30 raised recall from 0.533 to 0.733 and reduced missed risky cases from 42 to 24, at the cost of more false alarms (23 to 59), illustrating that calibration is a documentable governance decision. The study contributes a COSO-aligned architecture for explainability, a reproducible mapping protocol and empirical evidence on a public benchmark. Its limitations include the age and foreign origin of the data, the small sample, and the author-defined, non-validated feature-to-COSO mapping.

Keywords: explainable artificial intelligence; COSO framework; internal financial control; SHAP; AI governance; institutional sustainability

Highlights

- A four-layer framework links explainable AI outputs to the five COSO components.
- SHAP attributions on credit data are aggregated by COSO component.
- Interpretable logistic regression matches ensembles under repeated cross-validation.
- Decision-threshold choice is framed as a documented governance decision.
- Code and data are openly available for full reproducibility.

1. Introduction

Over the past decade, artificial intelligence (AI) has moved from experimental use to routine deployment in internal financial control, including fraud detection, credit-risk assessment and continuous auditing (Kokina & Davenport, 2017; Sun & Vasarhelyi, 2018). The best-performing models are often the least transparent, which makes it difficult for auditors and audit committees to reconstruct why an automated control flagged or cleared a transaction. This opacity sits uneasily with the principles of transparency, accountability and professional judgment on which control frameworks depend (Munoko et al., 2020; Zhang et al., 2022).

The COSO Internal Control–Integrated Framework (COSO, 2013) is the most widely adopted reference for designing and assessing internal control. It comprises five components: control environment, risk assessment, control activities, information and communication, and monitoring activities. The framework was conceived for controls operated by people, so integrating learning algorithms into it requires explicit mechanisms for explanation and accountability. Professional bodies have begun to issue guidance on applying enterprise risk-management and risk-governance frameworks to AI (COSO & Deloitte, 2021; NIST, 2023), but the explanation techniques themselves remain largely outside these frameworks.

Explainable AI (XAI) addresses the opacity problem through techniques such as Shapley-value attributions (Lundberg & Lee, 2017) and local surrogate models (Ribeiro et al., 2016); the literature on these techniques is extensive and well organized (Guidotti et al., 2018; Arrieta et al., 2020). However, to the author's knowledge, few studies translate quantitative XAI outputs into the vocabulary of a standard internal-control framework and test that translation empirically. This study addresses that gap.

The study makes three contributions. First, it proposes IFC-XAI, a four-layer governance framework whose third layer distributes explanation outputs across the five COSO components. Second, it specifies and applies a reproducible protocol for aggregating SHAP attributions by COSO component. Third, it reports empirical evidence on a public benchmark, including robustness checks (repeated cross-validation, bootstrap confidence intervals and a decision-threshold analysis) that qualify what the results can and cannot support.

The remainder of the paper is organized as follows. Section 2 reviews the background and related work. Section 3 describes the methodology. Section 4 presents the proposed framework. Section 5 reports the results. Section 6 discusses implications, Section 7 sets out limitations and future work, and Section 8 concludes.

2. Background and related work

2.1 The COSO internal control framework

COSO (2013) defines internal control as a process, effected by the board of directors, management and other personnel, designed to provide reasonable assurance regarding the achievement of objectives relating to operations, reporting and compliance. Table 1 summarizes its five components.

Table 1. The five components of the COSO Internal Control–Integrated Framework (COSO, 2013).

Component	Brief description
Control environment	Integrity, ethical values and the organizational structure that underpin control
Risk assessment	Identification and analysis of risks to the achievement of objectives
Control activities	Policies and procedures that ensure management directives are carried out
Information and communication	Timely exchange of relevant information internally and externally
Monitoring activities	Ongoing or periodic evaluation of whether the components are present and functioning

2.2 Explainable artificial intelligence

XAI refers to methods that make the decisions of complex models understandable to humans (Doshi-Velez & Kim, 2017). Surveys classify these methods by scope (global or local) and by dependence on

the model (model-specific or model-agnostic) (Adadi & Berrada, 2018; Arrieta et al., 2020). Two techniques are particularly prominent. SHAP is grounded in the Shapley value of cooperative game theory (Shapley, 1953); it allocates the difference between a prediction and the average prediction among the input features according to their marginal contributions (Lundberg & Lee, 2017), and efficient variants exist for tree ensembles (Lundberg et al., 2020). LIME approximates the behavior of a complex model around an individual prediction with a simple linear model (Ribeiro et al., 2016).

The field is not without criticism. Lipton (2018) argued that "interpretability" is used with divergent meanings and that claims about it require a precise statement of what is to be explained and to whom. Rudin (2019) contended that post-hoc explanations of opaque models can be unfaithful to the underlying model and that inherently interpretable models should be preferred for high-stakes decisions. The present study takes both critiques seriously: it uses an inherently interpretable model as its reference and employs SHAP to aggregate and map attributions to control components, rather than to justify an opaque model.

Explanation is also tied to regulatory accountability. Goodman and Flaxman (2017) discussed the transparency obligations that automated decision-making may entail under European regulation, and the NIST AI Risk Management Framework lists explainability and interpretability among the characteristics of trustworthy AI systems (NIST, 2023).

2.3 Related work in auditing and credit risk

In auditing, Kokina and Davenport (2017) examined how automation and AI are changing audit work, and Sun and Vasarhelyi (2018) illustrated the use of deep learning for textual data analytics in audit settings. Munoko et al. (2020) analyzed the ethical implications of AI in auditing and noted that a lack of transparency raises problems for accountability and for the auditor's professional judgment. Zhang et al. (2022) proposed a framework for applying XAI in auditing and identified research needs. In credit risk, Bussmann et al. (2021) applied a Shapley-value-based approach to explain credit-scoring models, supporting the feasibility of such explanations in financial settings.

At the governance level, COSO and Deloitte (2021) published guidance on applying the COSO enterprise risk management framework to AI, and NIST (2023) released a risk-management framework for AI systems. These works address, respectively, auditing practice, governance in general, and the technical explanation of models. In the literature reviewed informally by the author, the quantitative linkage between XAI outputs and the five components of the COSO internal-control framework received little empirical attention; this assessment rests on a non-systematic review and does not claim exhaustive coverage.

3. Methodology

The study follows the Design Science Research (DSR) paradigm (Hevner et al., 2004), which couples the construction of an artifact with its evaluation. Five activities were carried out: (i) problem identification; (ii) design of the conceptual framework; (iii) instantiation on real data; (iv) evaluation of predictive performance and explanations; and (v) discussion and formulation of recommendations.

3.1 Data

The German Credit dataset (Statlog) was obtained from the UCI Machine Learning Repository (Hofmann, 1994; Dua & Graff, 2019). It contains 1000 loan applicants described by 20 attributes (e.g., checking-account status, credit history, loan purpose, savings, employment duration, housing, property and age) and a binary label for credit outcome. The label was recoded so that 1 denotes credit risk (default; n = 300, 30.0%) and 0 denotes no risk (n = 700, 70.0%). Monetary values are expressed in

Deutsche Marks (DM), reflecting the data's origin. Table 2 reports descriptive statistics for the numeric attributes; none contains missing values.

Table 2. Descriptive statistics of the numeric attributes (n = 1000).

Variable	Mean	SD	Min	Median	Max
Duration (months)	20.90	12.06	4.0	18.0	72.0
Credit amount (DM)	3271.26	2822.74	250.0	2319.5	18424.0
Installment rate	2.97	1.12	1.0	3.0	4.0
Present residence (years)	2.85	1.10	1.0	3.0	4.0
Age (years)	35.55	11.38	19.0	33.0	75.0
Existing credits	1.41	0.58	1.0	1.0	4.0
People liable	1.16	0.36	1.0	1.0	2.0

3.2 Preprocessing and experimental design

Categorical attributes were one-hot encoded and numeric attributes standardized, yielding 61 features. Data were split into training (n = 700) and test (n = 300) sets using stratified sampling with a fixed random seed (42). Preprocessing was fitted within each training partition to avoid information leakage. All analyses used Python with scikit-learn 1.9.1 (Pedregosa et al., 2011), SHAP 0.53.0, NumPy 2.5.3, pandas 3.0.5 and SciPy 1.18.1.

3.3 Models and evaluation

Four classifiers were trained: logistic regression (Hosmer et al., 2013) as an inherently interpretable reference; a decision tree (maximum depth 6); a random forest (Breiman, 2001; 300 trees, maximum depth 8); and gradient boosting (Friedman, 2001; 200 estimators, maximum depth 3). Hyperparameters were fixed a priori and were not tuned, so the comparison reflects reasonable defaults and not optimized performance.

Performance was assessed on the hold-out test set using accuracy, precision, recall, F1-score, the area under the ROC curve (AUC; Fawcett, 2006) and the confusion matrix. Robustness was assessed through repeated stratified 5-fold cross-validation with three repeats (Kohavi, 1995), a paired t-test on fold-level AUC, a 95% bootstrap confidence interval for the logistic regression test AUC (2000 resamples; Efron & Tibshirani, 1993), and an analysis of decision thresholds between 0.20 and 0.50.

3.4 Explanation and mapping to COSO

SHAP attributions were computed for the logistic regression model using the linear explainer. For a linear model with independent features, the attribution of feature i for instance x is $\varphi_i(x) = \beta_i(x_i - E[x_i])$. The global importance of a feature was defined as the mean absolute attribution over the test set. The 15 most important encoded features were then assigned to COSO components using a functional classification defined by the author (Table 7). The explanatory weight of component c was computed as $\text{Share}(c) = \Sigma_{j \in c} \text{mean}|\varphi_j| / \Sigma_{j \in \text{top15}} \text{mean}|\varphi_j|$.

The classification is a heuristic and has not been validated by domain experts; it is intended to demonstrate the protocol, not to establish a definitive taxonomy. The assignment rule was as follows. Indicators of the applicant's financial standing and credit history were assigned to Risk Assessment. Parameters of the loan structure that a lending policy can bound (amount, duration, installment rate, number of existing credits) were assigned to Control Activities. Attributes of the applicant's personal

profile (personal status, employment duration, housing) were assigned to Control Environment, because they raise questions of policy, ethics and fairness that fall under the integrity and ethical-values principles of that component.

4. The proposed IFC-XAI framework

The framework rests on the premise that explainability should be designed as a bridge between an intelligent model and the governance system from the outset, and not applied as an afterthought. It comprises four successive layers connected by a continuous feedback loop (Fig. 1).

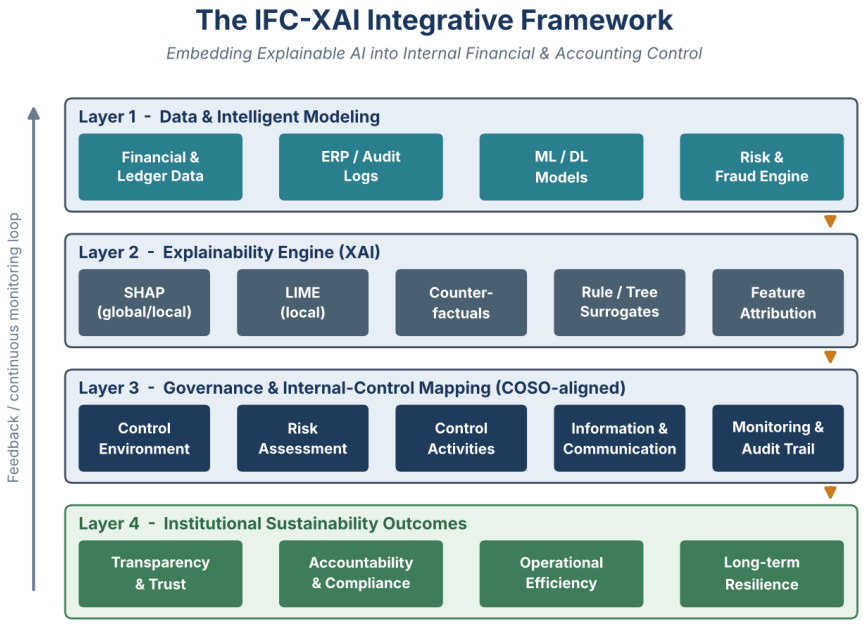


Fig. 1. The IFC-XAI framework for embedding explainable AI in internal financial and accounting control.

- Layer 1, Data and intelligent modeling: financial and ledger data, ERP and audit logs, ML/DL models, and risk and fraud engines.
- Layer 2, Explainability engine: SHAP, LIME, counterfactual explanations, rule- or tree-based surrogates and feature attribution, used as complementary techniques.
- Layer 3, Governance and internal-control mapping: distribution of explanation outputs across the five COSO components, which is the framework's principal conceptual contribution.
- Layer 4, Institutional sustainability outcomes: transparency and trust, accountability and compliance, operational efficiency, and long-term resilience.

The feedback loop makes the framework cyclical: results from monitoring and from the sustainability outcomes feed back into the data and models, so that a model is retrained or its explanation criteria revised when the risk pattern changes. Institutional sustainability is treated here as a conceptual outcome of the framework; it was not measured in the empirical part of this study.

4.1 Operationalizing Layer 3 through the five COSO components

Table 3. Role of explainability in each COSO component under IFC-XAI.

COSO component	Proposed role of XAI outputs
Control environment	An AI governance committee within the audit-committee structure approves models used in financial control and mandates a documented explanation (e.g., SHAP and LIME outputs) before activation.
Risk assessment	Machine-learning models classify financial risk, and the global SHAP analysis is attached to the annual risk-assessment document.
Control activities	Features with the highest SHAP weight (e.g., checking-account status, loan duration, installment rate) are tied to automated control limits that trigger alerts when exceeded.
Information and communication	A dashboard presents local SHAP explanations in plain language to finance managers and auditors, translating numeric attributions into understandable statements.
Monitoring activities	Periodic monitoring of the stability of feature importance over time; a material change in ranking signals a shift in the risk pattern or the need to recalibrate, and acts as an early-warning indicator.

Together, these roles turn the distribution of explanatory weight observed in the results (Section 5.4) into a deliberate functional integration, in which the outputs of Layer 2 feed information and communication and monitoring, so that explanation becomes part of a continuous control cycle.

5. Results

5.1 Predictive performance

Table 4 reports performance on the test set (n = 300) at the default threshold of 0.50. Logistic regression achieved the highest AUC (0.801), followed by random forest (0.797), gradient boosting (0.765) and the decision tree (0.633). The random forest obtained the highest precision (0.757) but the lowest recall (0.311), indicating a conservative classification of risky cases. The single decision tree performed worst, consistent with overfitting.

Table 4. Test-set performance of the four classifiers (threshold = 0.50).

Model	Accuracy	Precision	Recall	F1-score	AUC
Logistic Regression	0.7833	0.6761	0.5333	0.5963	0.8011
Random Forest	0.7633	0.7568	0.3111	0.4409	0.7971
Gradient Boosting	0.7633	0.6557	0.4444	0.5298	0.7651
Decision Tree	0.6567	0.4393	0.5222	0.4772	0.6326

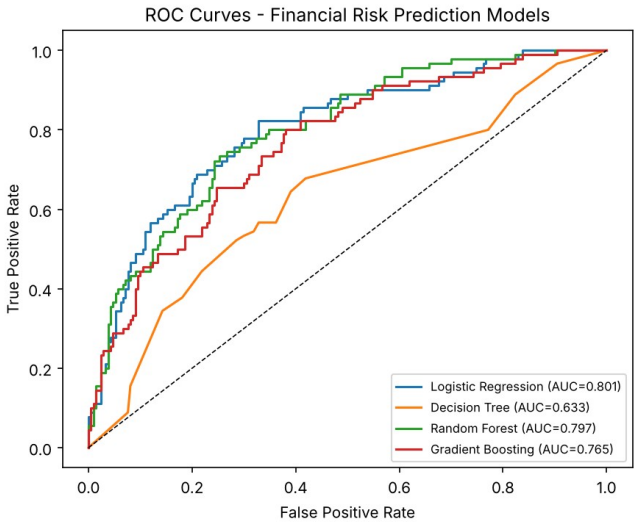


Fig. 2. ROC curves of the four classifiers on the test set.

The confusion matrix of the logistic regression model (Fig. 3) shows 187 true negatives and 48 true positives. Forty-two risky cases were classified as non-risky (type II errors), and 23 non-risky cases were classified as risky (type I errors). From an internal-control perspective, the type II error is typically the more costly, since it corresponds to a risk that the control fails to detect.

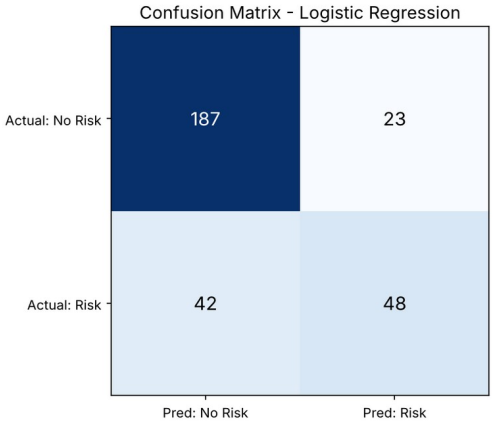


Fig. 3. Confusion matrix of the logistic regression model on the test set.

5.2 Robustness: cross-validation and confidence interval

Table 5 reports mean performance over 15 evaluations (5 folds × 3 repeats). Random forest (AUC = 0.7929) and logistic regression (AUC = 0.7806) were close, with random forest slightly higher on average. A paired t-test on fold-level AUC showed that the difference (0.0122) was not statistically significant at the 5% level ($t = -2.07$, $p = 0.058$). The advantage of logistic regression on the hold-out split should therefore not be interpreted as superiority. The 95% bootstrap confidence interval for the test AUC of logistic regression was 0.742 to 0.856, a wide interval reflecting the small test sample.

Table 5. Repeated stratified cross-validation results (5 folds × 3 repeats; mean values).

Model	Accuracy	Recall	F1-score	AUC	AUC SD
Logistic Regression	0.7497	0.4767	0.5334	0.7806	0.0309
Decision Tree	0.6987	0.3822	0.4240	0.6833	0.0407
Random Forest	0.7570	0.2911	0.4173	0.7929	0.0289
Gradient Boosting	0.7517	0.4878	0.5403	0.7750	0.0307

5.3 Decision-threshold analysis

Because recall at the default threshold was modest, the effect of lowering the threshold for the logistic regression model was examined (Table 6). At a threshold of 0.30, recall rose from 0.533 to 0.733 and missed risky cases fell from 42 to 24, while false alarms increased from 23 to 59. The appropriate operating point depends on the institution's relative cost of missed risks versus review workload; under IFC-XAI this choice is a documented management decision and not a default technical setting.

Table 6. Effect of the decision threshold on the logistic regression model (test set).

Threshold	Accuracy	Precision	Recall	F1	Missed risks (FN)	False alarms (FP)
0.5	0.7833	0.6761	0.5333	0.5963	42	23
0.4	0.7500	0.5773	0.6222	0.5989	34	41
0.3	0.7233	0.5280	0.7333	0.6140	24	59
0.25	0.7133	0.5141	0.8111	0.6293	17	69
0.2	0.6633	0.4663	0.8444	0.6008	14	87

5.4 SHAP attributions and mapping to COSO

Table 7 lists the 15 features with the largest mean absolute SHAP value, together with their assigned COSO component, and Fig. 4 shows the SHAP summary plot. The most influential features were the absence of a checking account (0.448), low savings (0.327), a critical credit history (0.322), the installment rate (0.317) and loan duration (0.285).

Table 7. Top 15 features by mean |SHAP| and their assigned COSO component.

Rank	Feature	Mean |SHAP|	COSO component
1	No checking account	0.448	Risk Assessment
2	Savings < 100 DM	0.327	Risk Assessment
3	Critical credit history / other credits existing	0.322	Risk Assessment
4	Installment rate	0.317	Control Activities
5	Loan duration (months)	0.285	Control Activities
6	Checking balance < 100 DM	0.228	Risk Assessment
7	Number of existing credits	0.206	Control Activities
8	Personal status: single male	0.199	Control Environment
9	Employment duration 4–7 years	0.183	Control Environment
10	No / unknown property	0.181	Risk Assessment
11	Housing: rent	0.167	Control Environment
12	Loan amount	0.166	Control Activities
13	No / unknown savings account	0.147	Risk Assessment
14	Checking balance 0–200 DM	0.139	Risk Assessment
15	Property: real estate	0.137	Risk Assessment

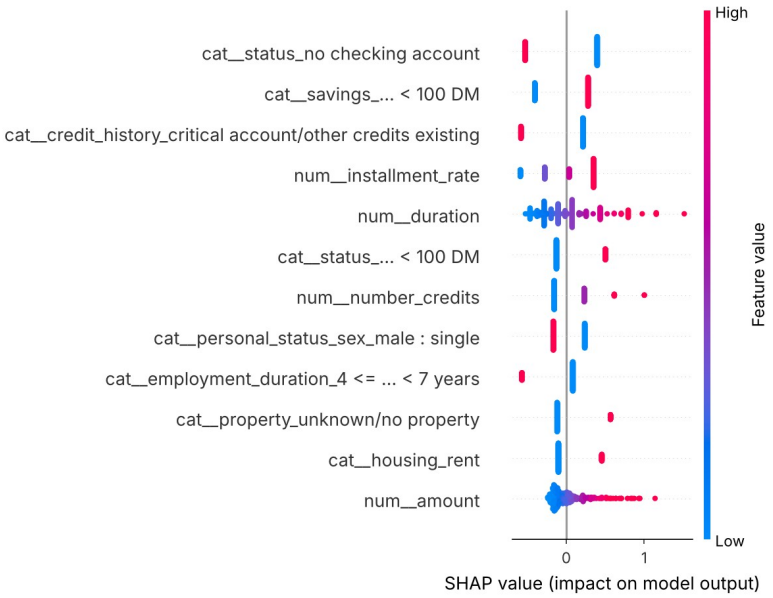


Fig. 4. SHAP summary plot of the most influential features (logistic regression).

Aggregating the weights by component gives the distribution in Table 8: Risk Assessment accounts for more than half of the explanatory weight (55.9%), followed by Control Activities (28.2%) and Control Environment (15.9%). Because the model is trained to predict risk, the dominance of Risk Assessment is expected; the analysis nonetheless shows that XAI can serve as a diagnostic instrument indicating where a control model's sensitivity is concentrated, and so where auditors' attention should be directed.

Information and communication and monitoring activities received no weight because they describe how explanations are used, and not input features; they are addressed in the framework's design (Table 3).

Table 8. Distribution of explanatory weight across COSO components.

COSO component	Features	Sum of mean |SHAP|	Share (%)
Risk Assessment	8	1.929	55.9
Control Activities	4	0.974	28.2
Control Environment	3	0.548	15.9

6. Discussion

The findings are consistent with the concern raised by Munoko et al. (2020) that opacity weakens the auditor's independent professional judgment: SHAP provides concrete, quantified evidence that can be examined and documented, which narrows that gap in practice. The statistical equivalence of the interpretable model and the ensemble models also supports the argument that the pursuit of maximum accuracy need not come at the expense of interpretability, in line with Rudin (2019). This holds only in the sense tested here: the difference between logistic regression and random forest was small and not significant on a sample of 1,000 cases, and the conclusion should not be extended to other data without further testing.

A practical implication is that institutional investment in XAI could be directed first toward risk assessment, where the largest share of explanatory weight resides in this case. A second implication concerns the threshold analysis: because the cost of a missed risk is usually higher than the cost of reviewing a false alarm, the choice of operating point is a governance decision that should be justified, approved and recorded, and XAI outputs can inform it.

A further observation concerns fairness. The personal-status attribute (single male) appears among the 15 most influential features. Using such attributes in credit decisions may be restricted by law or policy in many jurisdictions. That the explanation exposes it illustrates a concrete control use of XAI: surfacing sensitive or policy-relevant attributes for review under the control environment. The study does not assess discrimination formally, and this observation should be read as a prompt for review, not a finding of bias.

Recommendations for practice that follow from the framework are: (i) audit committees may require an XAI analysis to accompany any model used in control decisions; (ii) explanation tooling should prioritize the risk-assessment component; (iii) explanatory dashboards should be designed for non-technical users; (iv) the stability of feature importance should be monitored periodically as an early-warning signal; and (v) internal auditors should be trained to interpret SHAP and LIME outputs.

7. Limitations and future work

Several limitations qualify the results.

- Nature and age of the data. German Credit is a public credit dataset of German origin, donated to the UCI repository in 1994, with monetary values in Deutsche Marks. It does not represent an actual internal-control system, the Iraqi market or current credit conditions. It serves to test the framework, not to price loans or predict default in a local setting.
- Sample size. With 1,000 cases (300 in the test set), confidence intervals are wide and generalization is limited.

- COSO mapping. The feature-to-component assignment follows the author's functional classification and was not validated by auditors or control experts; different assignments could change the reported shares.
- Recall. Recall was 0.533 at the default threshold and improves only at the expense of more false alarms (Table 6).
- Model comparison. Differences between logistic regression and random forest are not statistically significant, and hyperparameters were not tuned.
- Explanation technique. Only SHAP was applied; it was not compared empirically with LIME or counterfactual explanations.
- Framework validation and sustainability. The framework has not been validated in the field with practitioners, and institutional sustainability was not measured empirically.

Future work includes applying IFC-XAI to real institutional control data; comparing XAI techniques on stability and consistency; validating the COSO mapping through expert panels; assessing fairness formally; and measuring the effect of explanatory dashboards on the quality of auditors' decisions in the field.

8. Conclusion

This study proposed IFC-XAI, a governance framework that integrates explainable AI with the COSO internal-control framework, and examined its feasibility on 1000 credit cases. An inherently interpretable model achieved performance comparable to more complex models (AUC = 0.801 on the hold-out set and 0.7806 under cross-validation), and SHAP provided a quantitative means of relating model behavior to control components, with Risk Assessment holding 55.9% of the explanatory weight. The threshold analysis showed that model calibration can be treated as a documented governance decision. The findings are limited by the nature of the data and the unvalidated mapping, and they should be tested on real institutional data before practical conclusions are drawn.

Declarations

CRediT authorship contribution statement. Ali Abdulwahab Yahya Al-Saffar: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of generative AI and AI-assisted technologies in the writing process. During the preparation of this work the author used Claude (Anthropic) to assist with drafting, code development, data analysis scripts and language editing. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication. [Author to confirm this statement matches the journal's policy and the extent of AI use.]

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Declaration of competing interest. [Author to confirm: the author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.]

Data and code availability. The German Credit dataset is publicly available from the UCI Machine Learning Repository. All code, derived tables and figures are available at https://github.com/AliAlsaffarMosulUniversity/ifc-xai-coso under the MIT license.

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